Computer Networks Part 1

DAY 60
08/09/2025

A Computer Network allows multiple computers/devices to connect and communicate with each other, Rather than working in isolation.

Without networks → Computers are like islands
With networks → they become part of a connected world

Why We Need Network?

1. Resource sharing
   - Share Hardware (printers, scanners, storage)
   - Share Software (licenced apps, centralized servers)
   - Save cost By avoiding duplication

2. Communication
   - Email, messaging, Video conferencing, VoIP.
   - Enables Collaboration
   - Foundation of remote work & social Media

3. Data & file sharing
   - share files instantly instead of physical transfer (USB, disk)
   - Access shared databases (banking, Hospital records, cloud storage)

4. Scalability
   - Easy to add new computers/users without major changes
   - Centralized management for large organizations.

5 Reliability and Backup
   - Distributed Systems (cloud computing, load Balancing)
   - if one server fails, other can take over)

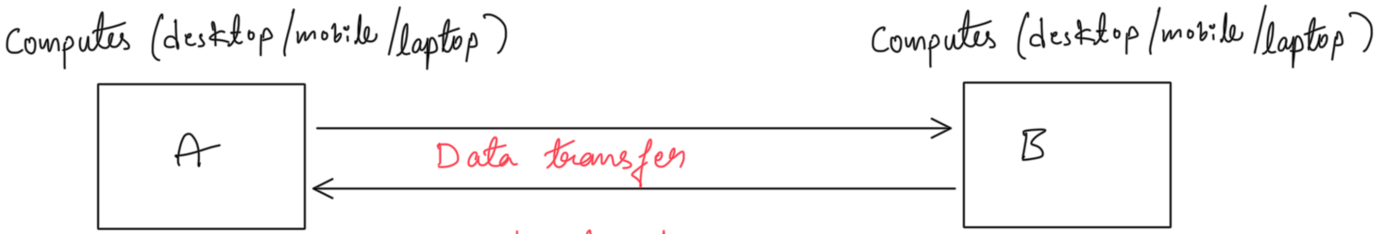
6 Remote Access

- Access your data or applications from anywhere (VPN, cloud, remote desktop)
- Enables IoT devices (Smart Home, Smart cities)

We need computer networks for connectivity, communication, sharing, and efficiency - they form the backbone of modern digital life.

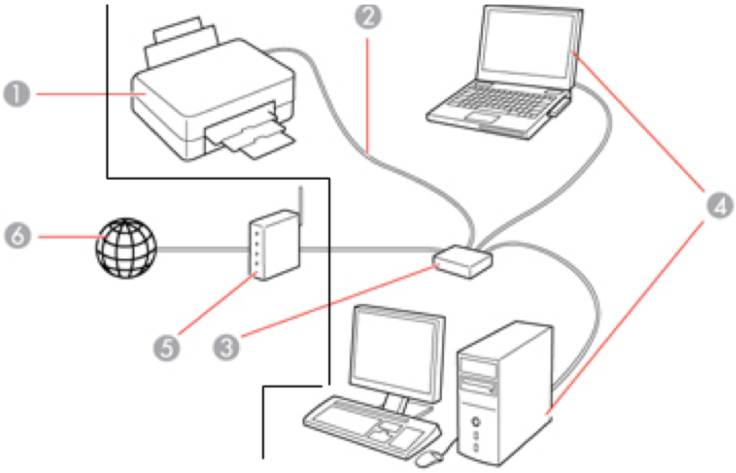
🔷 3. Real-Life Applications

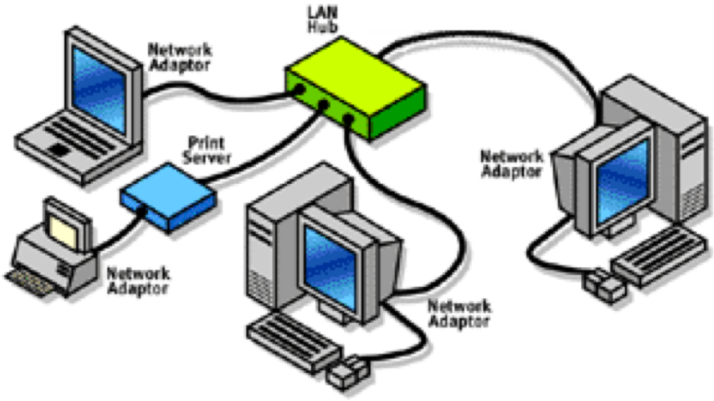
- 🌍 **Internet:** the largest network, connects billions of devices.
- 🏦 **Banking:** ATMs, online transactions, UPI.
- 🏥 **Healthcare:** patient records shared across hospitals.
- 🏢 **Corporate networks:** central servers, employee collaboration.
- 🚗 **Transportation:** GPS, ride-sharing apps.

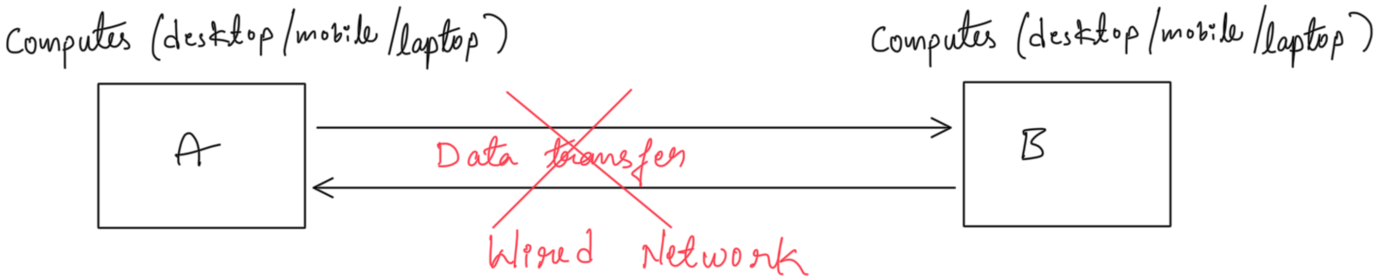


Wired Network

Before understanding Wireless Networks, understand the Wired Network. In the above figure two devices connected with a data transfer wire connection. Through which connection of two computers and data transfer happens.

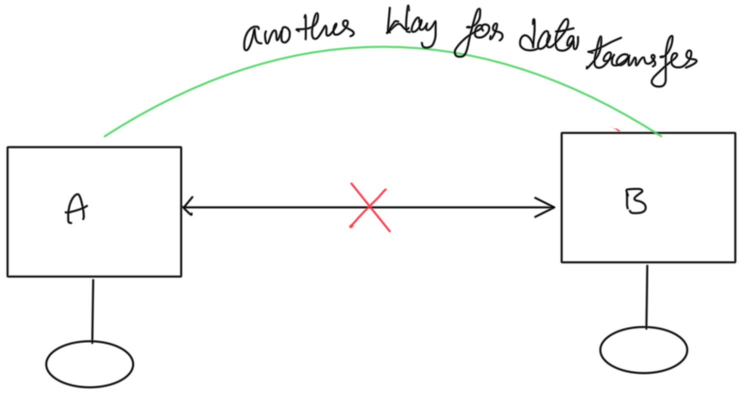


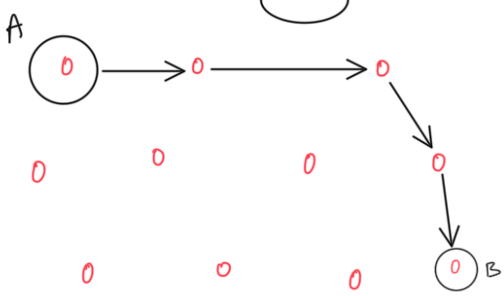




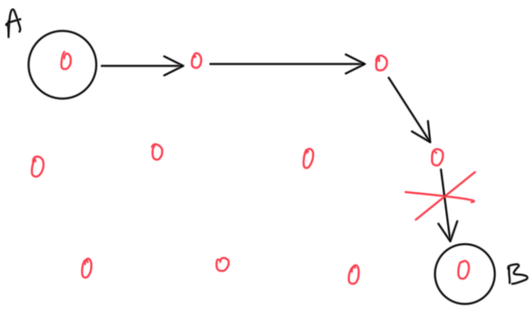
When the Wired Networks are Built to make Data transfer, But there is an issue with the Wired Network, like if the Cable Wired connection got cut then the communication between two computer stops.

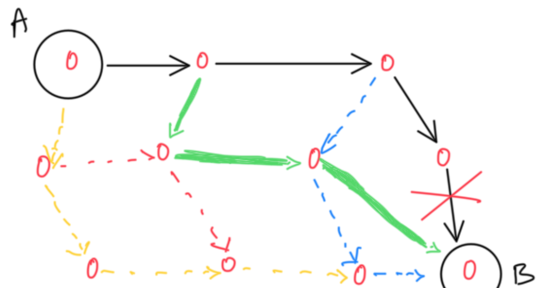
To overcome this issue Engineers wants the computers to find another way when a cable connection faces any failure





Imagine any one of the network cable connection collapsed in the group of computers which are connected together

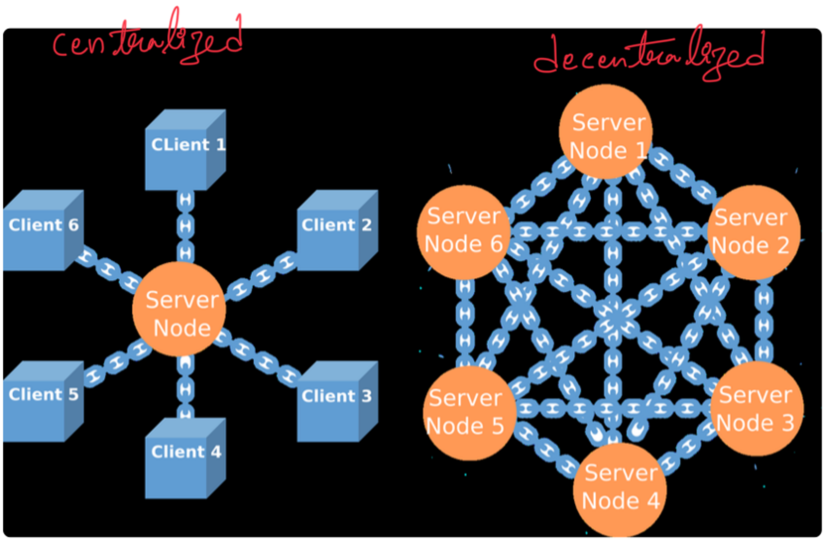




There must have others possible way for communication

We Just Have to decide destination from source the Network itself should find a way for data transfer.

So This was the Birth of New type of communication Called decentralized communication.

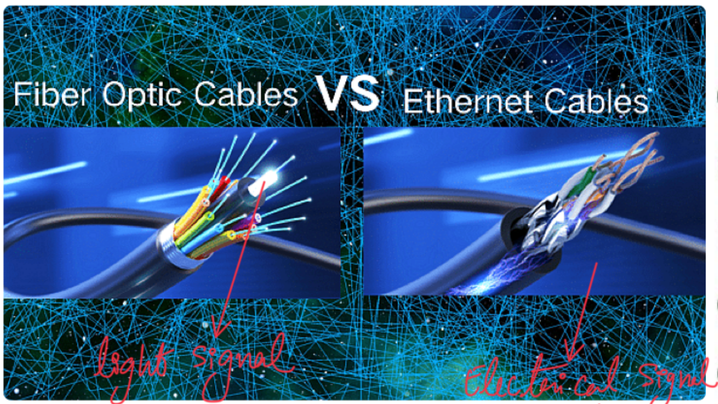


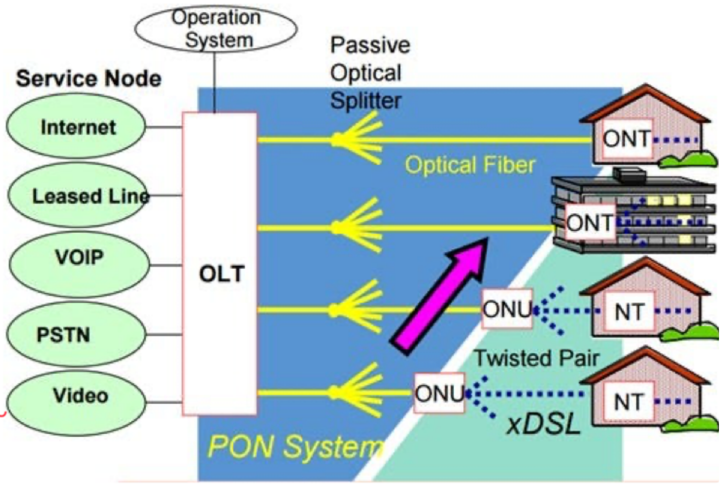
Feature	Centralized Network	Decentralized Network
Control	Single central server or authority	Shared across multiple nodes
Data Storage	Stored in one place	Distributed across many nodes
Single Point of Failure	Yes – if central server fails, entire network goes down	No – failure of one node doesn't stop the network
Performance	Fast for small scale (less communication overhead)	Can be slower due to coordination between nodes
Scalability	Limited – central server can become overloaded	High – load is distributed across nodes
Security	Easier to secure (only one main server to protect)	Harder – every node needs protection
Censorship / Control	Easy – central authority can block users/data	Difficult – no single authority
Examples	Gmail, Facebook, Banking systems	Blockchain, BitTorrent, Peer-to-Peer apps
Complexity	Simpler to design & maintain	More complex architecture & synchronization needed

Most of the Internet connections are using cable connection

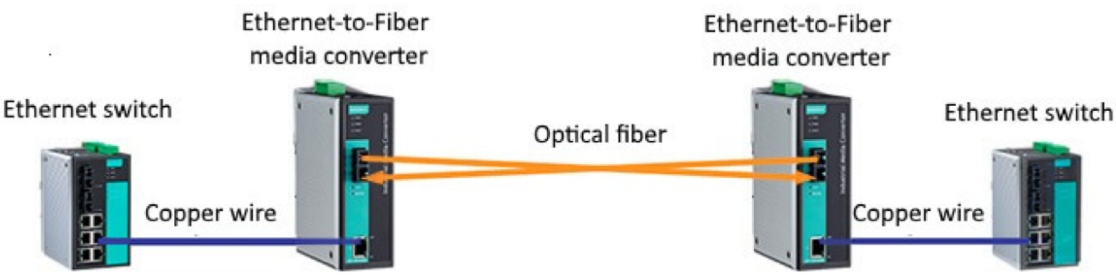
Before the modern Internet, computers were connected in **isolated wired networks (LANs)**. They faced many issues:

Problem in Old Wired Networks	How Internet Solved It
1. Limited Connectivity → Only computers within the same LAN/ building could communicate.	Internet connects **any device worldwide** via TCP/IP.
2. Proprietary Protocols → Each company used its own communication system (IBM SNA, Novell NetWare).	Internet standardized **TCP/IP** → universal language for all devices.
3. Expensive Point-to-Point Links → Direct dedicated wires were needed between locations.	Internet uses **packet switching** → multiple users share the same network efficiently.
4. Scalability Issues → Adding more devices required complex rewiring.	Internet is **scalable** — billions of devices connect dynamically via routers.
5. Reliability Issues → If one link failed, communication stopped.	Internet is **fault-tolerant** → packets can take alternate routes.
6. Limited Resource Sharing → Couldn't easily share files/printers globally.	Internet enables **global resource sharing** (cloud, remote storage, online apps).
7. Centralized Communication → Old networks often relied on a central machine (single point of failure).	Internet is **decentralized & distributed**, reducing dependency on one node.





But At the End Data is Converted in Binary format

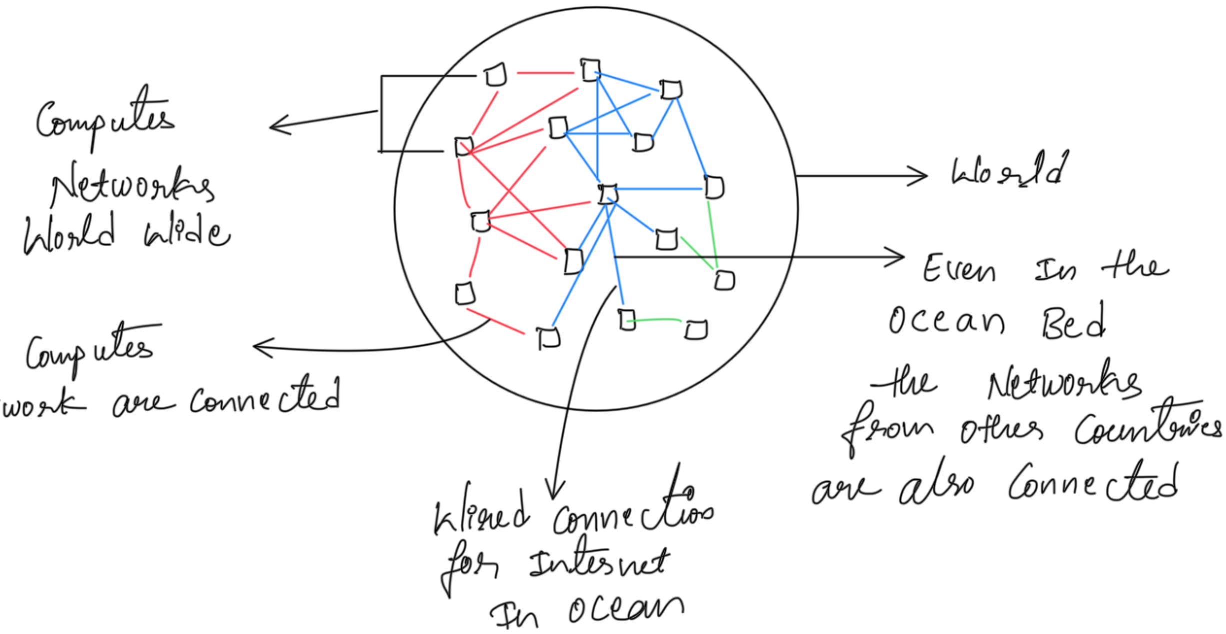


In Hardware level Information Always travels In Binary format. (1 or 0), and irrespective of what type of Information It is.

What Internet is :

The Internet is Basically a global System of Inter connected Networks that Communicate using TCP/IP protocol Suite

It is a network of networks, where your laptop, Phone, or IOT device can talk to any other device world wide.



Tomorrow I will Learn Indepth How Network or Internet Works